Preliminary conservation status and threats to Cyanopithecus scintillans (Primates: Cercopithecidae), Udzungwa Mountains, Tanzania

Zawadi Ngeleja[1*], Amani Kavalo[1], Nema Mseru[1], Elias R. Holloway[2]

[1]Department of Zoology and Wildlife Conservation, University of Dar es Salaam, Dar es Salaam, 35064, Tanzania.
[2]Department of Ecology and Evolution, Stony Brook University, Stony Brook, NY, 11794, USA.

*Corresponding author(s). E-mail(s): zawadi.ngeleja@udsm.ac.tz;

Abstract

Cyanopithecus scintillans (Bluebits) is a small nocturnal primate recently described from montane evergreen forest in the eastern Udzungwa Mountains, Tanzania, at 2,294–2,336 m a.s.l. The species is known from three naturally deceased specimens at a single locality; no population estimate or range-wide survey has been conducted. We present a preliminary conservation assessment based on the species description and regional literature, supplemented by spatial analyses of potential habitat, threat exposure, and elevational sensitivity. Seven patches of potential suitable habitat totalling 1,056 ha were identified across the eastern Udzungwas, 82% of which remains unsurveyed for nocturnal primates. Upward shifts of the suitable-elevation threshold by 100–300 m would reduce potential habitat by 42–93%. Minimum extent of occurrence (0.14 km^2) and area of occupancy (4–8 km^2) fall below Critically Endangered thresholds under IUCN Criterion B, though these values derive from tightly clustered records. Under precautionary scenarios the species could qualify as Vulnerable to Critically Endangered; Data Deficient remains defensible given current uncertainty. We recommend treating *C. scintillans* as an urgent conservation priority and conducting targeted nocturnal surveys of unsurveyed habitat, beginning with the Luhomero ridge.

Keywords: Conservation assessment, *Cyanopithecus scintillans*, Eastern Arc Mountains, IUCN Red List, montane forest

1 Introduction

1.1 Eastern Arc biodiversity and recent primate discoveries

The Eastern Arc Mountains of Tanzania and Kenya are among the most important biodiversity hotspots in the world, supporting exceptional levels of endemism across plants, invertebrates, and vertebrates [1]. Within this system, the Udzungwa Mountains of south-central Tanzania have proven especially productive for vertebrate discovery, yielding new species and genera of primates, sengis, and other mammals over the past two decades [2–4]. The continued description of new taxa from forests that have been subject to decades of biological survey underscores how incompletely the fauna of these mountains is known, particularly at higher elevations and for nocturnal or cryptic groups.

The discovery and subsequent conservation assessment of the kipunji *Rungwecebus kipunji* illustrates the trajectory that newly described primates in this region can follow. The species was described in 2005 as the first new African monkey genus in over eighty years [5]. Within three years, a systematic census and conservation status assessment estimated the total population at approximately 1,117 individuals and recommended Critically Endangered status on the IUCN Red List [6]. A reassessment thirteen years later demonstrated that sustained monitoring and conservation investment had stabilized and modestly improved the population [7]. This sequence — species description, rapid follow-up assessment, targeted conservation action, and long-term monitoring — provides a model for newly discovered taxa whose conservation status is likely to be precarious but whose baseline data are initially very limited. It should be noted, however, that the kipunji assessment was based on extensive distribution surveys, systematic census work, and quantitative population estimates; the present study does not include new field surveys or a population census, and the comparison is therefore procedural rather than evidentiary.

1.2 *Cyanopithecus scintillans* and scope of this assessment

Kavalo et al. [8] recently described *Cyanopithecus scintillans* gen. et sp. nov. (common name: Bluebits), a small nocturnal primate from montane evergreen forest in the upper Sanje corridor of the eastern Udzungwa Mountains, Tanzania. The species is provisionally placed in Cercopithecidae as sister to *Miopithecus* on the basis of multilocus phylogenetic analysis and is distinguished from all other cercopithecids by the combination of paired dermal photophores producing spontaneous visible bioluminescence, a semi-prehensile tail, enlarged pinnae and orbits, and strictly nocturnal habits. The description is based on three naturally deceased specimens and associated field observations from a single locality at 2,294–2,336 m a.s.l. No population estimate, distribution survey, or formal conservation assessment has been conducted.

Here we present a preliminary conservation status assessment of *C. scintillans*, following the kipunji model of rapid follow-up after species description. Our assessment is based entirely on published data from Kavalo et al. [8] and regional conservation literature; no new field data are introduced. We evaluate the available evidence against IUCN Red List criteria, identify and characterize threats using spatial analyses of

Table 1: Summary of survey effort and detections reported in the species description of *Cyanopithecus scintillans* [8]. All surveys were conducted in the eastern Udzungwa Mountains, Tanzania, in lower montane and submontane evergreen forest between 1,420 and 2,340 m a.s.l.

Parameter	Value	Unit
Survey effort		
Nocturnal visual surveys	61	person-hours
Camera trapping	214	trap-nights
Low-light video monitoring	96	trap-nights
Detections		
Specimens recovered (naturally deceased)	3	individuals
Observed group sizes	6–14	individuals per group
Confirmed locality records	3	sites (within ∼500 m)
Elevation range of records	2,294–2,336	m a.s.l.

habitat, fragmentation, and climate vulnerability, highlight critical data gaps, and outline research and conservation priorities. Our aim is to provide an evidence-based framework for conservation planning and to accelerate the path toward a formal IUCN assessment.

2 Methods

2.1 Data sources and IUCN framework

This assessment is based entirely on published data from the species description of *Cyanopithecus scintillans* [8], supplemented by regional conservation literature on the Eastern Arc Mountains and Udzungwa forests. No new field surveys were conducted. The evidence base from the original description comprises 61 person-hours of nocturnal visual surveys, 214 camera-trap nights, 96 low-light video trap-nights, and three naturally deceased specimens recovered from a single locality in the upper Sanje corridor (Table 1). This level of survey effort is sufficient to support a species description but is limited for drawing inferences about population size, distribution extent, or population trend.

We evaluated the conservation status of *C. scintillans* against the five IUCN Red List criteria (A–E; version 3.1) following the framework described by Mace et al. [9] and the current IUCN guidelines [10]. Criteria A (population reduction), C (small population size with decline), and E (quantitative extinction risk analysis) cannot be assessed because no population estimate, trend data, or population viability analysis exists for this species. We therefore focus on Criterion B (geographic range size) and Criterion D (very small or restricted population), for which the available locality and habitat data permit at least a preliminary evaluation (Section 5.1).

2.2 Habitat-envelope, threat-overlay, and elevational sensitivity analyses

To estimate the extent of potentially suitable habitat for *C. scintillans*, we delineated all patches of closed-canopy forest (≥70% tree cover; Hansen Global Forest Change v1.11) at elevations ≥2,200 m a.s.l. (SRTM 30 m DEM) across the eastern Udzungwa Mountains. Throughout this paper we distinguish three categories: *confirmed records* (the three specimen localities and associated observations from the type series), *minimum range metrics* (EOO and AOO calculated from confirmed records), and *potential suitable habitat* (the GIS-derived forest and elevation envelope). The habitat-envelope patches are analytical outputs and should not be interpreted as confirmed occupied range. Patches separated by more than 500 m of non-forest or sub-threshold elevation were treated as distinct units. Each patch was characterized by area, elevational range, protection status (World Database on Protected Areas, 2025), cumulative forest loss (Hansen GFC, 2001–2024), and proximity to settlements and roads (OpenStreetMap, 2026). A qualitative risk category (low, moderate, or high) was assigned to each patch by integrating forest loss rate, proximity to human access, and mountaintop limitation. Survey priority tiers were assigned using a heuristic scoring of elevation match, forest quality, proximity to known records, patch size, and threat urgency.

To evaluate exposure to climate-driven habitat loss, we calculated the total area of closed-canopy forest remaining above progressively higher elevation thresholds (2,000–2,550 m at 100 m intervals) across the eastern Udzungwas, simulating the effect of upslope contraction of suitable climatic conditions. This analysis assumes static forest cover and does not model upslope forest expansion.

3 Available evidence

3.1 Known records and minimum range metrics

Cyanopithecus scintillans is currently known from three locality records, all from the upper Sanje corridor in the eastern Udzungwa Mountains, within the Udzungwa Mountains National Park [8]. The three specimens were recovered at elevations of 2,294–2,336 m a.s.l. in closed-canopy montane evergreen forest, a habitat type characteristic of the upper elevational zones of the Udzungwa range [1, 4]. All records fall within approximately 500 m of one another, with a maximum inter-record distance of 0.76 km (Table 2).

The minimum convex polygon enclosing these records yields an extent of occurrence (EOO) of 0.14 km^2, and the IUCN-standard 2 km × 2 km grid produces an area of occupancy (AOO) of 4–8 km^2 depending on grid placement (Table 2). Both values fall far below the Critically Endangered thresholds under Criterion B (<100 km^2 EOO; <10 km^2 AOO). The species is presently assignable to a single IUCN-defined location, as all known records are subject to the same plausible threat regime. These range metrics should be interpreted as minimum estimates: the three records are tightly clustered and the resulting EOO is inherently unstable. The true distribution of *C. scintillans* may extend to other high-elevation forest patches in the eastern Udzungwas, but in the absence of targeted surveys this remains unknown.

Table 2: Distribution and range metrics for *Cyanopithecus scintillans*, derived from the three locality records reported in Kavalo et al. [8]. All records are from the upper Sanje corridor, Udzungwa Mountains National Park, Tanzania (approximately 07°48′S, 36°51′E). Extent of occurrence (EOO) is the minimum convex polygon area; area of occupancy (AOO) uses the IUCN-standard 2 km × 2 km grid. Values should be interpreted as lower bounds given the small number of tightly clustered records.

Metric	Value	Notes
Locality records	3	All within upper Sanje corridor
Elevational range of records	2,294–2,336 m	42 m span
Maximum inter-record distance	0.76 km	
Minimum EOO (convex hull)	0.14 km^2	Below CR threshold (<100 km^2)
Minimum AOO (2 km grid)	4–8 km^2	1–2 occupied grid cells
Number of locations (IUCN)	1	Single threat-defined location

3.2 Population and natural history

No population estimate exists for *Cyanopithecus scintillans*. The type series comprises three naturally deceased specimens, and no inference about population size should be drawn from this number. Camera-trap records and nocturnal visual observations reported group sizes of 6–14 individuals during the original surveys [8], but the total number of groups, the degree of re-observation of the same individuals, and the extent of the survey area relative to the species' range are all unknown. The available data are therefore insufficient to estimate even an order-of-magnitude population size.

Several natural history traits reported by Kavalo et al. [8] are relevant to conservation planning. *Cyanopithecus scintillans* is strictly nocturnal, a trait rare among cercopithecids and one that substantially reduces detectability during conventional diurnal primate surveys. This raises the possibility that the species has been overlooked by previous survey efforts in the Udzungwa Mountains, which have focused predominantly on diurnal taxa [4, 11]. The species was observed in closed-canopy montane evergreen forest on low branches, moss-covered logs, and the forest floor, consistent with use of structurally complex understory microhabitats at high elevation. Repeated clicking vocalizations and a low-frequency hum were associated with group movement, indicating that acoustic monitoring may offer a viable survey approach for future population assessment. The bioluminescent display itself, although of considerable biological interest, has no established relevance to threat assessment at this stage.

4 Threat assessment

4.1 Habitat loss, degradation, fragmentation, and local disturbance

The Eastern Arc Mountains have experienced substantial forest loss over the past century, driven by agricultural expansion, logging, charcoal production, and infrastructure development [12, 13]. Although rates of deforestation within formally protected areas

Table 3: Potential habitat patches for *C. scintillans* in the eastern Udzungwa Mountains, with threat exposure and survey priority. Patches defined as closed-canopy forest (≥70% tree cover) at elevations ≥2,200 m, separated by >500 m of non-forest or sub-threshold elevation. Forest loss from Hansen GFC (2001–2024). Risk categories (L/M/H) integrate forest loss rate, proximity to human access, and mountaintop limitation (elevation ceiling constraining upslope retreat). Survey priority tiers are heuristic, based on elevation match, forest quality, proximity to known records, patch size, and threat urgency (Tier 1 = highest). UMNP = Udzungwa Mountains National Park; KNR = Kilombero Nature Reserve.

ID	Patch	Elev. (m)	Forest loss (%)	Risk	Survey	Area (ha)
P1	Upper Sanje (UMNP)	2,200–2,386	2.4	M	Confirmed	186
P2	Luhomero ridge (UMNP)	2,200–2,576	1.1	L	Tier 1	412
P3	Luhomero W. (UMNP)	2,200–2,348	3.8	M	Tier 2	94
P4	Ndundulu (KNR)	2,200–2,468	1.9	M	Tier 2	238
P5	Nyumbanitu (KNR)	2,200–2,312	5.6	H	Tier 2	67
P6	Mwanihana summit (UMNP)	2,200–2,264	7.2	H	Tier 3	41
P7	Iwonde fragment	2,200–2,241	12.1	H	Tier 3	18
Total						**1,056**
Within protected areas						971 (92%)
Unsurveyed for nocturnal primates						870 (82%)

have been lower than in surrounding landscapes, degradation and edge effects continue to erode forest quality even within gazetted reserves [1, 14]. Within the Udzungwa Mountains specifically, hunting pressure and habitat degradation have been identified as primary drivers of primate population declines [14].

Using the habitat-envelope approach described in Section 2.2, we identified seven closed-canopy forest patches at elevations ≥2,200 m across the eastern Udzungwa Mountains, totalling 1,056 ha of potential suitable habitat (Table 3). The largest patch, Luhomero ridge (P2; 412 ha), encompasses the highest point in the Udzungwa range (2,576 m) and retains the lowest rate of forest loss (1.1%) among all patches. The confirmed locality of *C. scintillans*, the upper Sanje corridor (P1; 186 ha), represents the second-largest patch and falls entirely within the Udzungwa Mountains National Park. Four additional patches occur within the national park or the Kilombero Nature Reserve, bringing the proportion of potential habitat under formal protection to 92%. A single small fragment (P7, Iwonde; 18 ha) lies outside any protected area and has experienced the highest rate of forest loss (12.1%).

Despite the high level of formal protection, the total area of potential habitat is small and fragmented across seven discrete patches. The three smallest and most isolated patches (P5–P7) are classified as high risk on the basis of elevated forest loss rates, proximity to human settlements, and severe mountaintop limitation, which restricts the capacity for upslope retreat under warming conditions. Critically, 82% of the total potential habitat (870 ha) has never been surveyed for nocturnal primates. Whether *C. scintillans* occupies any patch beyond the confirmed type locality remains unknown.

The fragmented distribution of suitable habitat raises concerns about population connectivity. The closest unsurveyed patch to the type locality, Luhomero ridge (P2), lies approximately 3.2 km edge-to-edge from P1, separated by degraded montane forest at elevations as low as 2,080 m. More distant patches (P4–P7) are separated by 5–25 km of mixed forest, grassland, or cultivation mosaic at substantially lower elevations (1,520–1,780 m). For a small-bodied nocturnal primate with no documented terrestrial dispersal behavior, these gaps may represent significant barriers to gene flow and recolonization, although dispersal capacity in *C. scintillans* is entirely unknown.

Hunting and human disturbance represent additional potential threats. Although no hunting of *C. scintillans* has been documented, hunting pressure on primates is widespread across the Udzungwa Mountains [14], and the Wahehe hunters who provided early accounts of luminous mammals in the Kilombero Valley [8] indicate that human activity extends into high-elevation forest. The three high-risk patches (P5–P7) lie within 4 km of settlements, increasing exposure to hunting, firewood collection, and other forms of disturbance [15].

4.2 Climate change and elevational range contraction

Climate change poses a particular threat to tropical montane species, which are expected to track suitable conditions upslope as temperatures rise [16]. Because land area decreases with elevation on any mountain, upslope shifts progressively compress the available habitat until species are stranded on summits with nowhere further to retreat. Empirical evidence for this process is accumulating: upslope range shifts and mountaintop extirpations have been documented in tropical bird communities [17], including in the Afrotropical montane forests of the Eastern Arc system itself [18]. Climate projections for Tanzanian forests indicate continued warming and potential contraction of montane forest zones over the coming decades [19].

Cyanopithecus scintillans is known from a narrow elevational band (2,294–2,336 m) near the upper limit of closed-canopy forest in the eastern Udzungwa Mountains. Using the elevational sensitivity analysis described in Section 2.2, we calculated the total area of closed-canopy forest remaining above progressively higher elevation thresholds across the eastern Udzungwas (Table 4).

The results reveal a steep decline in potential suitable habitat with increasing elevation. At the current habitat-envelope baseline ($\geq$2,200 m), 1,056 ha of potentially suitable forested habitat are distributed across seven patches. A 100 m upslope shift of suitable conditions reduces this to 614 ha across five patches, a loss of 41.9%. A 200 m shift leaves only 268 ha in three patches (-74.6%), and a 300 m shift reduces habitat to 72 ha confined to a single patch on the Luhomero ridge (-93.2%). Near the summit ($\geq$2,550 m), only 18 ha remain.

The steepness of this decline reflects the conical geometry of the Udzungwa range: small increases in the lower elevation limit eliminate disproportionately large areas of habitat. For *C. scintillans*, which is already restricted to high-elevation forest, even modest upslope contraction would substantially reduce the total area of potentially suitable habitat, eliminate the smaller and more isolated patches first, and confine remaining potential habitat to the Luhomero ridge. This analysis assumes static forest cover and does not account for potential upslope forest expansion, which observational

Table 4: Elevation-area analysis of forested habitat (≥70% canopy cover) in the eastern Udzungwa Mountains. Each row shows the total area of closed-canopy forest above the specified elevation threshold, simulating upslope contraction of suitable climatic conditions. Percentage loss is relative to the ≥2,200 m baseline. Known records of *C. scintillans* span 2,294–2,336 m a.s.l. Analysis assumes static forest cover and does not account for potential upslope forest expansion, which is unlikely to keep pace with projected warming rates in tropical montane systems [16, 17].

Threshold (m)	Forested area (ha)	Patches	Loss vs. baseline	Scenario
≥2,000	2,840	11	—	Extended buffer
≥2,100	1,924	9	—	Lower montane limit
≥2,200	1,056	7	0%	**Baseline (current envelope)**
≥2,300	614	5	−41.9%	+100 m shift
≥2,400	268	3	−74.6%	+200 m shift
≥2,500	72	1	−93.2%	+300 m shift
≥2,550	18	1	−98.3%	Near summit

and modeling evidence suggests is unlikely to keep pace with the rate of warming in tropical montane systems [16, 17]. The combination of a narrow known elevational range, limited summit area, and fragmented habitat distribution makes *C. scintillans* highly vulnerable to climate-driven range contraction.

4.3 Small population risks and compounding threats

The available evidence does not permit estimation of total population size for *C. scintillans*. However, the clustering of all confirmed records within a single small high-elevation habitat patch (P1; 186 ha) means that, if the known locality represents a substantial fraction of the species' range, the species could be exposed to the demographic and genetic risks typical of small, isolated populations [20, 21]. These risks include elevated vulnerability to demographic stochasticity, environmental variation, and loss of genetic diversity through drift and inbreeding. They are compounded when populations are isolated, as reduced gene flow accelerates the fixation of deleterious alleles and limits adaptive potential. Whether the population of *C. scintillans* is in fact small enough to face these pressures cannot be determined from the available data, but the restricted extent of confirmed records and the limited area of potential suitable habitat make this a plausible concern.

The threats identified in Sections 4.1 and 4.2 — habitat fragmentation, climate-driven range contraction, and potential exposure to hunting and disturbance — do not act in isolation. For a species whose confirmed records are restricted to a single high-elevation locality, these pressures could interact in ways that amplify overall risk. Habitat loss reduces carrying capacity; climate-driven upslope contraction eliminates potential habitat patches, increasing isolation; and isolation in turn heightens genetic vulnerability. Each individual threat carries considerable uncertainty for *C. scintillans*, but the convergence of multiple risk factors on the same species in the same

landscape raises the overall probability of decline. This pattern of correlated vulnerability is characteristic of narrow-range montane endemics and is a recognized driver of extinction risk even when no single threat can be quantified with precision [22].

5 Preliminary IUCN Red List assessment

5.1 Assessable and non-assessable criteria

Three of the five IUCN Red List criteria cannot be evaluated for *C. scintillans* given current data. Criterion A (population reduction) requires evidence of a past, present, or projected decline in population size, which is unavailable for a species described from three specimens with no baseline census. Criterion C (small population size with continuing decline) requires both an estimate of the number of mature individuals and evidence of decline or extreme fluctuation, neither of which can be established. Criterion E (quantitative analysis) requires a population viability analysis or equivalent model, which cannot be constructed without demographic parameters. These criteria may become assessable as survey and monitoring data accumulate (Section 6.1).

Criterion B (geographic range) is the most relevant criterion given available data, but its application must be treated with caution. The minimum EOO of 0.14 km^2 and minimum AOO of 4–8 km^2 (Table 2) both fall well below the Critically Endangered thresholds of 100 km^2 and 10 km^2, respectively, and the species is known from a single IUCN-defined location. However, these values are derived from only three tightly clustered records and are likely to change substantially as new data emerge. If the known records approximate the true range, the species could qualify as Critically Endangered under B1ab(iii) + B2ab(iii): sub-criterion (a) would be supported by the fragmented distribution of suitable habitat across seven isolated patches (Table 3), and sub-criterion b(iii) by documented forest loss [12, 13] and projected climate-driven contraction of high-elevation forest (Table 4). If future surveys reveal a broader distribution encompassing multiple habitat patches, the assessment could shift to Endangered or Vulnerable under the same criterion. The current range metrics should therefore be read as a lower bound on a provisional assessment, not as a definitive evaluation.

Criterion D is partially assessable. No population estimate exists, so sub-criteria D and D1 (fewer than 50 or 250 mature individuals) cannot be formally evaluated. However, Criterion D2 applies to species with an AOO below 20 km^2 or fewer than five locations, provided a plausible future threat exists that could drive the species to Critically Endangered or Extinct in a very short time. With an AOO of 4–8 km^2, a single known location, and documented threats including habitat degradation and climate-driven range contraction, *C. scintillans* meets the D2 threshold for Vulnerable independently of any population estimate.

5.2 Preliminary listing recommendation

To make the dependence of the assessment on current data limitations transparent, we evaluated the IUCN criteria under five scenarios representing different assumptions about the distribution, population size, and trend of *C. scintillans* (Table 5). These

Table 5: Preliminary IUCN Red List assessment for *Cyanopithecus scintillans* under five scenarios. Criteria follow IUCN version 3.1. Sub-criteria: a = severely fragmented or ≤5 locations; b(iii) = continuing decline in habitat quality; b(v) = continuing decline in mature individuals. D2: AOO <20 km^2 or ≤5 locations with plausible threat capable of driving rapid decline. NE = not evaluable; VU = Vulnerable; EN = Endangered; CR = Critically Endangered; DD = Data Deficient. Under most precautionary scenarios the species qualifies for at least Vulnerable; available evidence favors EN–CR. If no continuing decline can be inferred, Criterion B sub-criteria are insufficient and the assessment may fall to VU under D2 alone or to DD. A provisional threatened listing is favored over Data Deficient, though the latter remains defensible under formal IUCN treatment given the limited number of records [9, 22].

Scenario	Crit. B	Crit. D	A, C, E	Category
S1. Known records only; no trend data	CR: B1ab(iii) + B2ab(iii)	VU: D2	NE	**CR**
S2. Occupies high-suitability patches (P1–P4); no trend data	EN: B1ab(iii)	VU: D2	NE	**EN**
S3. Known records only; continuing habitat decline inferred	CR: B1ab(iii,v) + B2ab(iii,v)	VU: D2	NE	**CR**
S4. Occupies all patches (P1–P7); >250 mature individuals	VU: B1ab(iii)	—	NE	**VU**
S5. Known records only; no evidence of continuing decline	NE (sub-criteria insufficient)	VU: D2	NE	**VU or DD**

range from the most conservative interpretation of available evidence (S1: species confirmed only from known records, no trend data) to the most optimistic (S4: species occupies all seven identified habitat patches and exceeds 250 mature individuals).

Under most precautionary scenarios, *C. scintillans* would qualify for a threatened category. Under the assumptions most consistent with current evidence — a species known from a single locality, with an EOO of 0.14 km^2, an AOO of 4–8 km^2, and one IUCN-defined location (S1, S3) — the species could qualify as Critically Endangered under Criterion B1ab(iii) + B2ab(iii), provided the known records approximate the true range. Even if the species is eventually found to occupy all high-suitability patches (S2), it would likely still qualify as Endangered. Only under the most optimistic scenario (S4), in which the species occupies all seven patches with a population exceeding 250 mature individuals, does the assessment fall to Vulnerable. However, if no continuing decline can be inferred (S5), Criterion B sub-criteria are insufficient and the assessment rests on Criterion D2 alone, yielding Vulnerable at best or Data Deficient if the uncertainty is judged too great. The plausible category range is therefore Data Deficient to Critically Endangered, with the weight of current evidence favoring Endangered or Critically Endangered under the precautionary assumption that habitat decline is ongoing.

We recognize that a formal IUCN assessment could reasonably conclude Data Deficient, particularly if no continuing decline in habitat quality can be inferred, in which case Criterion B cannot be triggered and the assessment would rest on D2 alone. However, we note that under most precautionary scenarios the species qualifies for at least

a Vulnerable listing, and that assigning Data Deficient when available evidence points toward elevated risk can delay conservation action for species that are in fact threatened [22]. The IUCN framework encourages assessors to use the best available evidence rather than defaulting to Data Deficient when uncertainty, though substantial, does not prevent a plausible assessment [9]. We therefore favor a provisional threatened listing over Data Deficient, while acknowledging that either treatment could be defended under formal IUCN procedures until more data are available.

The kipunji provides an instructive precedent. Jones et al. [5] anticipated that *Rungwecebus kipunji* would be classified as Critically Endangered at the time of its description, and this was confirmed three years later by the first systematic census [6]. A subsequent reassessment thirteen years on demonstrated that sustained monitoring and conservation investment could stabilize and even modestly improve the status of a critically small population [7]. Unlike the kipunji assessment, however, the present study does not include new field surveys or a population census; the parallel is in the urgency of rapid follow-up, not in the strength of the underlying data. We recommend that *C. scintillans* be treated as provisionally Critically Endangered or Endangered under Criterion B pending targeted surveys to refine the range, population size, and trend data needed for a formal IUCN submission.

6 Research priorities and conservation recommendations

6.1 Survey and monitoring needs

The most urgent priority is to determine whether *C. scintillans* occurs beyond the type locality. The habitat-envelope analysis identified six unsurveyed potential habitat patches above 2,200 m in the eastern Udzungwa Mountains (Table 3), of which the Luhomero ridge (P2) should receive highest priority given its large area, proximity to the confirmed site, and intact forest cover. Surveys should employ methods suited to a small-bodied nocturnal primate: nocturnal transect walks with headlamps or thermal optics, camera traps with infrared flash deployed on low branches and fallen logs, and passive acoustic recorders targeting the clicking vocalizations and low-frequency hum described by Kavalo et al. [8]. The documented bioluminescent display may also offer a non-invasive detection cue under dark-adapted conditions, although its reliability for survey purposes has not been tested. Diurnal primate survey protocols, which form the basis of most existing Udzungwa monitoring programs [11, 23], are unlikely to detect this species and may explain its absence from previous inventories [4].

Once the distribution is better resolved, systematic population estimation should follow, using repeated surveys or spatially explicit capture–recapture methods adapted from comparable small-primate systems. Baseline population and demographic data are prerequisites for evaluating IUCN Criteria C and D (Section 5.1) and for designing a long-term monitoring framework. The kipunji program provides a useful methodological model, although the scale of survey effort required for *C. scintillans* will need to be adapted to a nocturnal, small-bodied species: Davenport et al. [6] established a complete-count protocol within three years of the species description, enabling a

subsequent reassessment thirteen years later that demonstrated measurable conservation outcomes [7]. Habitat mapping should be refined using high-resolution satellite imagery and ground-truthing to characterize forest structure, canopy condition, and connectivity between patches. Ecological and behavioral studies — activity budgets, diet, ranging patterns, group structure, and reproductive biology — are needed to inform habitat management. Genetic sampling from non-invasively collected material (hair, feces) would allow assessment of population structure and diversity without requiring capture, and should be framed to support conservation planning rather than taxonomic revision [21].

6.2 Immediate conservation actions and path to formal IUCN assessment

Conservation action should not await the completion of these research programs. The precautionary principle supports immediate protection of all identified potential habitat patches, particularly those within existing protected areas where management measures can be strengthened without new legislative action. The type locality lies within the Udzungwa Mountains National Park, and integration of *C. scintillans* into the park's species monitoring and management plans should be straightforward. For the Kilombero Nature Reserve patches (P4, P5), coordination with reserve management to include nocturnal primate monitoring in routine patrol and survey activities would extend coverage at minimal additional cost.

Community engagement offers both conservation and survey benefits. The Wahehe term *Viumbe vya Bluu* and associated hunter accounts of luminous mammals in the Kilombero Valley [8] demonstrate existing local ecological knowledge that could guide survey design and identify additional localities. Participatory monitoring programs involving local communities have proven effective for other rare primates in the region [14] and could be adapted for *C. scintillans*. Survey effort should also extend beyond the Udzungwa Mountains to other Eastern Arc blocks with high-elevation montane forest, including the Uluguru and Rubeho ranges, to test whether the species' distribution is broader than presently known.

To progress from this preliminary assessment to a formal IUCN Red List submission, the following minimum evidence package is needed: confirmed range limits based on systematic surveys of all suitable habitat, a defensible population estimate with confidence bounds, documentation of specific threats and their severity at occupied sites, and at least two temporally separated assessments to establish population trend [9, 10]. Until these data are available, we recommend that *C. scintillans* be treated as provisionally Critically Endangered or Endangered and managed accordingly, while recognizing that a formal IUCN assessment may conclude Data Deficient if the uncertainty surrounding the species' range is judged too great for a definitive listing.

7 Conclusion

Cyanopithecus scintillans is presently known from three specimens and a single locality in the upper Sanje corridor of the eastern Udzungwa Mountains. Despite these

limited data, the available evidence consistently points toward elevated extinction risk. The species' known range metrics fall below the Critically Endangered thresholds under IUCN Criterion B, though these values are derived from only three clustered records and may not reflect the true range. The total area of potentially suitable high-elevation forest is small and fragmented, and climate-driven upslope contraction could eliminate the majority of this habitat within decades. Under most precautionary scenarios examined here, the species would qualify for a threatened category, and the weight of current evidence supports a provisional assessment of Critically Endangered or Endangered, with Data Deficient remaining a defensible alternative under formal IUCN treatment until the species' distribution is better resolved.

Substantial uncertainty remains. The true distribution, population size, and ecology of *C. scintillans* are essentially unknown, and the preliminary nature of this assessment must be emphasized. However, uncertainty is not grounds for inaction. The kipunji precedent demonstrates that rapid follow-up assessment and targeted conservation investment after species description can secure measurable outcomes, even when initial data are limited. Targeted nocturnal surveys of the six unsurveyed potential habitat patches identified here, beginning with the Luhomero ridge, represent the most urgent next step. Until these data are available, precautionary protection of all identified potential habitat, integration into existing protected area management, and engagement with local communities should proceed without delay.

References

[1] Burgess, N.D., Butynski, T.M., Cordeiro, N.J., Doggart, N., Fjeldså, J., Howell, K., Kilahama, F.B., Loader, S., Lovett, J.C., Mbilinyi, B., *et al.*: The biological importance of the eastern arc mountains of tanzania and kenya. Biological conservation **134**(2), 209–231 (2007)

[2] Dinesen, L., Lehmberg, T., Rahner, M.C., Fjeldså, J.: Conservation priorities for the forests of the udzungwa mountains, tanzania, based on primates, duikers and birds. Biological Conservation **99**(2), 223–236 (2001)

[3] Rovero, F., Rathbun, G.B., Perkin, A., Jones, T., Ribble, D.O., Leonard, C., Mwakisoma, R., Doggart, N.: A new species of giant sengi or elephant-shrew (genus rhynchocyon) highlights the exceptional biodiversity of the udzungwa mountains of tanzania. Journal of Zoology **274**(2), 126–133 (2008)

[4] Rovero, F., Marshall, A.R., Jones, T., Perkin, A.: The primates of the udzungwa mountains: diversity, ecology and conservation. Journal of Anthropological Sciences= Rivista Di Antropologia: JASS **87**, 93–126 (2009)

[5] Jones, T., Ehardt, C.L., Butynski, T.M., Davenport, T.R.B., Mpunga, N.E., Machaga, S.J., Luca, D.W.D.: The highland mangabey ¡i¿lophocebus kipunji¡/i¿: A new species of african monkey. Science **308**(5725), 1161–1164 (2005) https://doi.org/10.1126/science.1109191 https://www.science.org/doi/pdf/10.1126/science.1109191

[6] Davenport, T.R., De Luca, D.W., Jones, T., Mpunga, N.E., Machaga, S.J., Kitegile, A., Phillipps, G.P.: The critically endangered kipunji rungwecebus kipunji of southern tanzania: first census and conservation status assessment. Oryx **42**(3), 352–359 (2008)

[7] Davenport, T.R., Machaga, S.J., Mpunga, N.E., Kimiti, S.P., Mwalwengele, W., Mwaipungu, O., Makumbule, P.M.: A reassessment of the population size, demography, and status of tanzania's endemic kipunji rungwecebus kipunji 13 years on: Demonstrating conservation success. International Journal of Primatology **43**(2), 317–338 (2022)

[8] Kavalo, A., Mseru, N., Holloway, E.R.: A new bioluminescent primate genus and species (Primates: Cercopithecidae) of the Udzungwa Mountains, Tanzania. Zenodo (2026). https://doi.org/10.5281/zenodo.20025918 . https://doi.org/10.5281/zenodo.20025918

[9] Mace, G.M., Collar, N.J., Gaston, K.J., Hilton-Taylor, C., Akçakaya, H.R., Leader-Williams, N., Milner-Gulland, E.J., Stuart, S.N.: Quantification of extinction risk: Iucn's system for classifying threatened species. Conservation biology **22**(6), 1424–1442 (2008)

[10] (2025). https://www.iucnredlist.org/resources/categories-and-criteria

[11] Rovero, F., Struhsaker, T.T., Marshall, A.R., Rinne, T.A., Pedersen, U.B., Butynski, T.M., Ehardt, C.L., Mtui, A.S.: Abundance of diurnal primates in mwanihana forest, udzungwa mountains, tanzania. International Journal of Primatology **27**(3), 675–697 (2006)

[12] Hall, J., Burgess, N.D., Lovett, J., Mbilinyi, B., Gereau, R.E.: Conservation implications of deforestation across an elevational gradient in the eastern arc mountains, tanzania. Biological conservation **142**(11), 2510–2521 (2009)

[13] Green, J.M., Larrosa, C., Burgess, N.D., Balmford, A., Johnston, A., Mbilinyi, B.P., Platts, P.J., Coad, L.: Deforestation in an african biodiversity hotspot: Extent, variation and the effectiveness of protected areas. Biological Conservation **164**, 62–72 (2013)

[14] Rovero, F., Mtui, A.S., Kitegile, A.S., Nielsen, M.R.: Hunting or habitat degradation? decline of primate populations in udzungwa mountains, tanzania: An analysis of threats. Biological conservation **146**(1), 89–96 (2012)

[15] Ehardt, C.L., Jones, T.P., Butynski, T.M.: Protective status, ecology and strategies for improving conservation of cercocebus sanjei in the udzungwa mountains, tanzania. International Journal of Primatology **26**(3), 557–583 (2005)

[16] Colwell, R.K., Brehm, G., Cardelus, C.L., Gilman, A.C., Longino, J.T.: Global warming, elevational range shifts, and lowland biotic attrition in the wet tropics.

science **322**(5899), 258–261 (2008)

[17] Freeman, B.G., Scholer, M.N., Ruiz-Gutierrez, V., Fitzpatrick, J.W.: Climate change causes upslope shifts and mountaintop extirpations in a tropical bird community. Proceedings of the National Academy of Sciences **115**(47), 11982–11987 (2018)

[18] Neate-Clegg, M.H., Stuart, S.N., Mtui, D., Şekercioğlu, Ç.H., Newmark, W.D.: Afrotropical montane birds experience upslope shifts and range contractions along a fragmented elevational gradient in response to global warming. PLoS One **16**(3), 0248712 (2021)

[19] John, E., Bunting, P., Hardy, A., Roberts, O., Giliba, R., Silayo, D.S.: Modelling the impact of climate change on tanzanian forests. Diversity and Distributions **26**(12), 1663–1686 (2020)

[20] Shaffer, M.L.: Minimum population sizes for species conservation. BioScience **31**(2), 131–134 (1981)

[21] Frankham, R.: Genetics and extinction. Biological conservation **126**(2), 131–140 (2005)

[22] Borgelt, J., Dorber, M., Høiberg, M.A., Verones, F.: More than half of data deficient species predicted to be threatened by extinction. Communications biology **5**(1), 679 (2022)

[23] Plumptre, A.J., Cox, D.: Counting primates for conservation: primate surveys in uganda. Primates **47**(1), 65–73 (2006)